# PAPER_723: UQFF Quantum Variable Documents: gamma, E_react, f_quasi, R_b

**Class:** `UQFFKnowledgeBaseKB9` **CP4 Entry:** #307 **Keywords:** UQFF, gamma decay, E_react, reaction energy, f_quasi, quasi-static, R_b, buoyancy boundary, omega_c **Session:** 176 | **Version:** v5.33 **Source:** grok_share_ba508f76c8e.txt

## Abstract

Five quantum variable documents define: $\gamma$ (UQFF damping rate for magnetic string oscillations), $E_{react}(t)$ (reaction energy expressing the vacuum energy available for field driving), $f_{quasi}$ (quasi-static field perturbation), $R_b$ (buoyancy Heaviside step radius), and $\omega_c$ (UQFF slow-cycle frequency coupling atomic to galactic timescales).

## 1. Variable Registry

| Variable | Value | Physical Role |
|---|---|---|
| $\gamma$ | $5 \times 10^{-5}$ day$^{-1}$ | Damping in $e^{-\gamma t \cos(\pi t_n)}$ |
| $E_{react}(0)$ | $10^{46}$ J | Vacuum energy at UQFF scale |
| $f_{quasi}$ | 0.01 | Perturbation: $(1 + f_{quasi})$ in $U_m, U_H$ |
| $R_b$ | $3.086 \times 10^{19}$ m (1 kpc) | Heaviside step for U_g2 profile |
| $\omega_c$ | $1.585 \times 10^{-8}$ rad/s | Slow-cycle: $\mu_j(t)\sin(\omega_c t)$ |

## 2. Reaction Energy Decay

$$E_{react}(t) = 10^{46} \cdot e^{-\kappa t} \quad (\kappa = 5 \times 10^{-4}\,\text{day}^{-1})$$

At 100 years: $E_{react}(36500) \approx 1.61 \times 10^{44}$ J.

## 3. Buoyancy Step Profile

$$U_{g2}(r) = k_2 \cdot \frac{(\rho_{UA} + \rho_{SCm})M_s}{r^2} \cdot S(r - R_b) \cdot (1 + \delta_{sw}v_{sw}) \cdot H_{SCm} \cdot E_{react}$$

where $S(r - R_b)$ is the Heaviside step activating at $r = R_b = 1$ kpc.

## References

- UQFF KB grok_share_ba508f76c8e.txt entry #73
- Session 176, v5.33

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